

IME 775 – Lecture 5

Quadratic Forms, Positive Definiteness, and Matrix Norms

Chapter 4: Finding Structure in Data

Core Challenge: High-dimensional data often lies near lower-dimensional subspaces.

Tools we'll learn:

- Quadratic forms → understand loss surfaces
 - Positive definiteness → convex optimization
 - Matrix norms → measure approximation quality
 - PCA & SVD → dimensionality reduction
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Quadratic Forms: Definition

Definition: For symmetric matrix A , the quadratic form is:

$$Q(\vec{x}) = \vec{x}^T A \vec{x}$$

For 2D with $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$:

$$Q = ax_0^2 + 2bx_0x_1 + cx_1^2$$

Workout: Expand Q for $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$:

Solution:

Using $Q = ax_0^2 + 2bx_0x_1 + cx_1^2$ with $a = 3, b = 1, c = 2$:

$$Q(\vec{x}) = \begin{bmatrix} x_0 & x_1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = 3x_0^2 + 2x_0x_1 + 2x_1^2$$

Geometric Meaning: Conic Sections

Quadratic forms describe **conic sections**:

Equation	Shape
$(\vec{x} - \vec{a})^T \mathbf{I} (\vec{x} - \vec{a}) = r^2$	Circle
$(\vec{x} - \vec{a})^T A (\vec{x} - \vec{a}) = 1$	Ellipse

Key Insight: Level sets of Q are ellipses (or hyperbolas) whose axes align with eigenvectors!

Quadratic Forms in ML

Where quadratic forms appear:

1. **Taylor expansion:** $L(\vec{x} + \delta) \approx L(\vec{x}) + \nabla L \cdot \delta + \frac{1}{2} \delta^T H \delta$
 2. **Squared error:** $\|\vec{y} - \hat{\vec{y}}\|^2 = (\vec{y} - \hat{\vec{y}})^T (\vec{y} - \hat{\vec{y}})$
 3. **Regularization:** $\|\vec{w}\|^2 = \vec{w}^T \vec{w}$
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Extrema on the Unit Sphere

Question: For unit vectors $\hat{x}$ (with $\|\hat{x}\| = 1$), what are the max/min of Q ?

Key Trick: Use spectral decomposition $A = S \Lambda S^T$:

$$Q = \hat{x}^T A \hat{x} = \hat{y}^T \Lambda \hat{y} = \sum_{i=1}^n \lambda_i y_i^2$$

where $\hat{y} = S^T \hat{x}$ is also a unit vector.

Theorem: Eigenvalues = Extrema

Since $\sum_i y_i^2 = 1$ and $y_i^2 \geq 0$:

$$Q = \sum_i \lambda_i y_i^2$$

is a **weighted average** of eigenvalues.

Result:

- **Maximum:** $Q_{max} = \lambda_{max}$ when $\hat{x} = \vec{e}_1$ (largest eigenvector)
- **Minimum:** $Q_{min} = \lambda_{min}$ when $\hat{x} = \vec{e}_n$ (smallest eigenvector)

Workout: If $\lambda_1 = 5$, $\lambda_2 = 2$, what are max/min of Q on unit circle?

Solution:

By the extrema theorem, on the unit sphere the quadratic form $Q = \sum_i \lambda_i y_i^2$ is maximized when all weight is on the largest eigenvalue and minimized when all weight is on the smallest:

$$Q_{max} = \lambda_1 = 5 \quad (\text{achieved at eigenvector } \vec{e}_1)$$

$$Q_{min} = \lambda_2 = 2 \quad (\text{achieved at eigenvector } \vec{e}_2)$$

Positive Definite Matrices

Definition: Symmetric A is:

- **Positive Definite (PD):** $\vec{x}^T A \vec{x} > 0$ for all $\vec{x} \neq 0$
- **Positive Semidefinite (PSD):** $\vec{x}^T A \vec{x} \geq 0$ for all $\vec{x}$

Eigenvalue Test:

$$A \text{ is PD} \iff \text{all } \lambda_i > 0$$

$$A \text{ is PSD} \iff \text{all } \lambda_i \geq 0$$

Why Positive Definiteness Matters

ML Applications:

1. **Covariance matrices** are always PSD
2. **Hessian at minimum** must be PD (second derivative test)
3. **Convex losses** have PSD Hessians everywhere

Workout: Is $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ positive definite? (Find eigenvalues)

Solution:

Characteristic equation: $\det(A - \lambda I) = (2 - \lambda)^2 - 1 = 0$

$$\lambda^2 - 4\lambda + 3 = 0 \implies (\lambda - 3)(\lambda - 1) = 0$$

$$\lambda_1 = 3, \quad \lambda_2 = 1$$

Both eigenvalues are strictly positive, so A is **positive definite**.

Indefinite Matrices: Saddle Points

If A has both positive and negative eigenvalues:

- A is **indefinite**
- Level sets are **hyperbolas**
- Critical point is a **saddle point**

Workout: Classify $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$:

Eigenvalues: $\det(A - \lambda I) = 0$

Solution:

$$(1 - \lambda)^2 - 4 = 0 \implies \lambda^2 - 2\lambda - 3 = 0 \implies (\lambda - 3)(\lambda + 1) = 0$$

$$\lambda_1 = 3 > 0, \quad \lambda_2 = -1 < 0$$

One positive and one negative eigenvalue $\rightarrow A$ is **indefinite**. The critical point of $Q(\vec{x}) = \vec{x}^T A \vec{x}$ is a **saddle point**, and the level sets are hyperbolas.

Condition Number

Definition: For positive definite A :

$$\kappa(A) = \frac{\lambda_{max}}{\lambda_{min}}$$

Interpretation:

- $\kappa \approx 1$: circular level sets $\rightarrow$ fast convergence
- $\kappa \gg 1$: elongated ellipses $\rightarrow$ slow convergence (zig-zagging)

Ideal: $\kappa = 1$ (all eigenvalues equal)

Spectral Norm

Definition: Maximum amplification by matrix A :

$$\|A\|_2 = \max_{\|\hat{x}\|=1} \|A\hat{x}\|$$

Result: Spectral norm = largest singular value:

$$\|A\|_2 = \sigma_1 = \sqrt{\lambda_{\max}(A^T A)}$$

ML Use: Lipschitz constant of linear layer, spectral normalization.

Computing Spectral Norm

For unit vector $\hat{x}$:

$$\|A\hat{x}\|^2 = (A\hat{x})^T(A\hat{x}) = \hat{x}^T A^T A \hat{x}$$

This is a quadratic form in $A^T A$!

Maximum occurs along eigenvector of $A^T A$ with largest eigenvalue.

Workout: Find $\|A\|_2$ for $A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$:

Solution:

A is diagonal, so its singular values equal the absolute values of its diagonal entries: $\sigma_1 = 3, \sigma_2 = 2$.

$$\|A\|_2 = \sigma_1 = 3$$

Equivalently: $A^T A = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix}$, so $\lambda_{\max}(A^T A) = 9$ and $\|A\|_2 = \sqrt{9} = 3$.

Frobenius Norm

Definition: "Size" of a matrix (like L2 for vectors):

$$\|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2}$$

Also equals:

$$\|A\|_F = \sqrt{\sum_i \sigma_i^2} = \sqrt{\text{trace}(A^T A)}$$

ML Use: Regularization, approximation error.

Workout: Compute $\|A\|_F$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$:

Solution:

$$\|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2} = \sqrt{1^2 + 2^2 + 3^2 + 4^2} = \sqrt{1 + 4 + 9 + 16} = \sqrt{30} \approx 5.477$$

Norm Comparison

Always: $\|A\|_2 \leq \|A\|_F$

Equality when: A has rank 1.

Norm	Formula	Measures
Spectral	σ_1	Max stretch
Frobenius	$\sqrt{\sum \sigma_i^2}$	Total "size"

Unit Circle Transformation

When A acts on unit circle:

- Circle maps to **ellipse**
- Semi-axes = singular values σ_1, σ_2
- Spectral norm = longest semi-axis

Workout: Sketch how $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ transforms unit circle:

Solution:

A is diagonal with singular values $\sigma_1 = 2, \sigma_2 = 1$.

The unit circle $x_0^2 + x_1^2 = 1$ maps to an **ellipse** with semi-axis length 2 along x_0 and semi-axis length 1 along x_1 :

$$\frac{x_0^2}{4} + x_1^2 = 1$$

The x_0 direction is stretched by factor 2, while the x_1 direction is unchanged. The spectral norm is $\|A\|_2 = 2$ (the longest semi-axis).

Summary: Key Formulas

Concept	Formula
Quadratic Form	$Q = \vec{x}^T A \vec{x}$
Max/Min on sphere	$\lambda_{max}, \lambda_{min}$
PD Test	All $\lambda_i > 0$
Condition Number	$\kappa = \lambda_{max} / \lambda_{min}$
Spectral Norm	$\ A\ _2 = \sigma_1$
Frobenius Norm	$\ A\ _F = \sqrt{\sum \sigma_i^2}$